



Integrated Design of a Controller and Compliant Power Supply for USB Power Delivery

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OBJECTIVE

The primary objectives of this project are to design and implement a USB PD 3.1 compliant controller, and a USB PD 3.1 compliant power supply using BCD technology. Additionally, the project aims to integrate these components and evaluate the performance metrics of the designed power supply

DESIGN

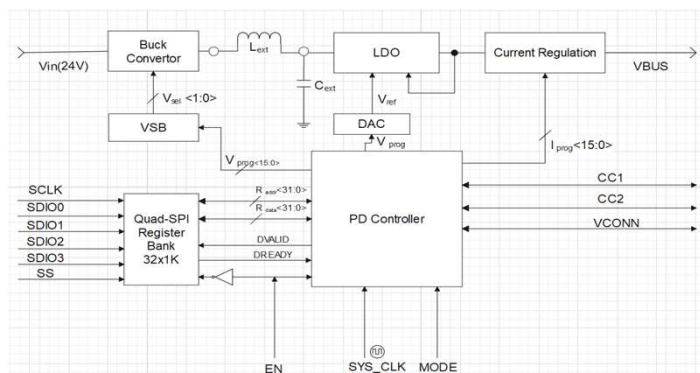


Fig 1: Architecture of the proposed design

IMPLEMENTATION

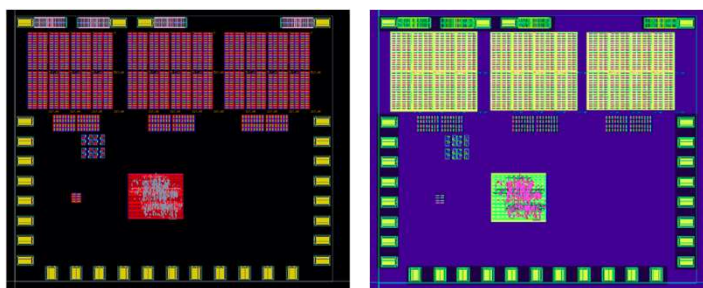


Fig 2: Layout and EMIR views of Integrated Chip

RESULTS

Power Delivery Controller			Programmable Power Supply		
Parameter	Specification	Result	Parameter	Specification	Result
Nominal Voltage	1.2V	1.2V	Operating Input Voltage	24V	24V
Total Power consumption	< 1mW	0.5114mW	Maximum Output Power	100W	100W
Std. Cell Count	< 10K	5502	Total Dissipated Power	< 2W	8.0025W
Clock frequency	100MHz	100MHz	Switching frequency	40KHz	40KHz
Die Area	< 1mm ²	0.7563mm ²	Die Area	< 25mm ²	4.3122mm ²
			Max. Regulated Current O/P	5A	5.2A
			Output Voltages	5V, 9V, 15V, 20V	5V, 9V, 15V, 20V
			Total Consumed Power	-	110.3W

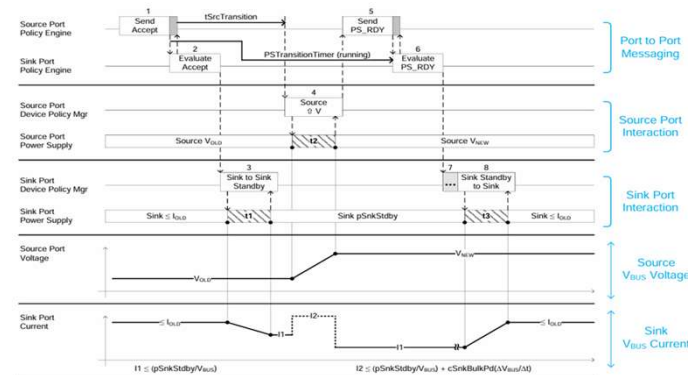


Fig 3: Voltage Transition Flow

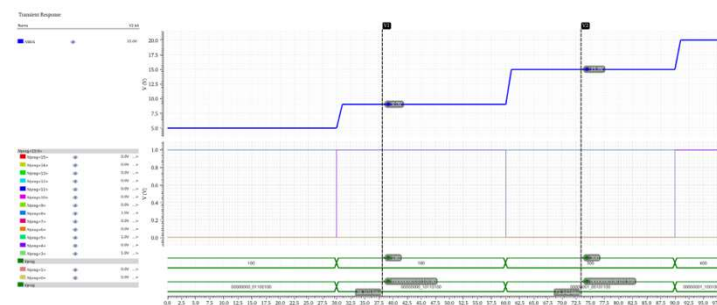


Fig 4: Vprog signal vs VBUS

Follow Up

- Integrate the current design with USB 4 PHY and Bus Controller.
- Qualify the design for Automotive Grade ($T_j = 150^{\circ}\text{C}$)

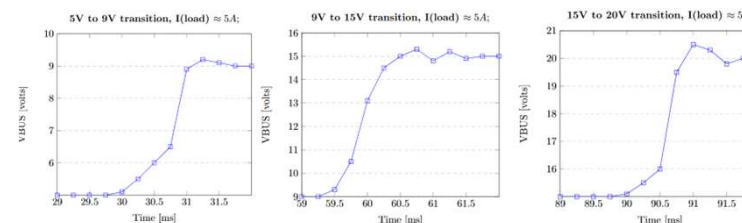


Fig 5: Voltage Transitions

Conclusions

- The proposed solution reduces the expense from BOM by integrating Power Delivery Controller and Power Supply .
- The effectiveness of LDMOS at 100W negates the need of Gallium Nitride(GaN) devices which are 4 x more expensive.

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